Analytical Chemistry Cumulative Exam
September 24, 2021

Topic: Fundamental Electrochemistry and Electrochemical Techniques

Attached at the end of this Cume is an equation sheet with useful and not-so-useful information. ***Show your work.***

1. **(40 points)** Consider the following electrochemical cell at 25 °C, written in shorthand notation:

   Pb | $PbSO_4$ *(s)* | SO_4^{2-} (1.00 M) || H^+ (2.00 M), SO_4^{2-} (1.00 M) | $PbSO_4$ *(s)* | PbO_2 *(s)*

   Where…
   $PbSO_4$ *(s)* + 2 e^- = Pb*(s)* + SO_4^{2-}*(aq)* **$E°_{Pb/PbSO4}$ = -0.355 V**
   PbO_2 *(s)* + 4 H^+ *(aq)* + SO_4^{2-}*(aq)* + 2 e^- = $PbSO_4$ *(s)* + 2 H_2O*(l)* **$E°_{PbO2/PbSO4}$ =+1.685 V**

   (a) **Identify which half reaction** occurs at the **anode** and which one occurs at the **cathode**.

   (b) Calculate the **potentials of the half-reactions**, given the concentrations above.

   (c) What are the **oxidation numbers** (oxidation states) of the lead in all three of the leadcontaining species ($PbSO_4$, Pb, and PbO_2).

   (d) Write the balanced, overall chemical reaction of the cell.

   (e) In the overall chemical reaction, which one of the reactants serves as the **oxidant**?

   (f) **Calculate the potential of the cell** (E_{cell}) and state whether the cell is **galvanic, electrolytic, or neither**.

   (g) Is the overall cell reaction **spontaneous or not**?

   (Just a point of interest: this is the chemistry involved in a lead-acid battery, which doesn't actually need a salt bridge.)

2. **(60 points) Electrochemical Techniques.** Select one of the following instrumental techniques used for electrochemical methods of analysis and answer each of the lettered questions about your chosen technique.

<u>Select one electrochemical technique:</u>
Chronopotentiometry
Chronoamperometry
Chronocoulometry
Linear Sweep Voltammetry
Cyclic Voltammetry
Square Wave Voltammetry
Differential Pulse Voltammetry
Electrochemical Impedance Spectroscopy

<u>Answer all of these questions about your the one technique that you have selected:</u>

(a) Does this technique involve applying a current or applying a potential?

(b) Draw a diagram showing the applied waveform (of the applied current or applied potential) as a function of time for a typical experiment or part of a typical experiment involving this technique. Identify any experimental conditions or restrictions on the chemistry in the experiment that would make the response true. (Be sure to label the axes)

(c) Draw a plot showing a typical electrochemical response for this technique. (Be sure to label the axes). Explain why the electrochemical response looks the way it does–how the applied potential or current produces the electrochemical response?

(d) Explain how the electrochemical response in (c) is affected by the following changes, ***if at all***. Try to be as quantitative as possible. Draw the expected electrochemical response resulting from this change if that will help you in your explanation. *(Note any special conditions that must be met, if any, for your answer to be true, in case there are a variety of possible answers or different ways to interpret the question)*

i. Doubling the concentration of the electroactive analyte.
ii. Doubling the diffusion coefficient of the electroactive analyte.
iii. Doubling the heterogeneous (electron transfer) rate constant of the electroactive analyte.
iv. Doubling the time scale of the experiment (experiment takes twice as long to complete).
v. Doubling the amplitude of the applied potential or applied current.
vi. Doubling the number of electrons transferred per electroactive analyte molecule
vii. Doubling the area of the electrode.
viii. Doubling the E° value (standard reduction potential) for the half reaction of the electroactive analyte

Analytical Cumulative Exam
Some Equations and Constants

Some Constants

$R = 0.08205746(14)$ L atm K^{-1} mol^{-1} = $8.3144621(75)$ J K^{-1} mol^{-1}

$F = 96485.3$ Coul/mol e^-

$RT/F = 0.02569$ V at 25 °C

$f = F/RT = 38.92$ V^{-1} at 25 °C

$e = 1.60218 \times 10^{-19}$ C

$\varepsilon_0 = 8.85419 \times 10^{-12}$ C^2 N^{-1} m^{-2} or F/m

k (Boltzmann constant) = 1.38065×10^{-23} J/K

Some equations that might or might not be helpful

$$E_{rxn} = E_{cell} = E_{cathode} - E_{anode} = E_{right} - E_{left} = E_{red} - E_{ox}$$

E_X vs. new ref. = E_X vs. old ref. - $E_{\text{new ref.}}$ vs. old ref.

$$E = E^{o} + \frac{RT}{nF} \ln \prod a^{\nu_i} = E^{o} + \frac{RT}{nF} \ln \frac{(O)}{(R)}$$

$$E = E^{o'} + \frac{RT}{nF} \ln \prod [x]^{\nu_i} = E^{o'} + \frac{RT}{nF} \ln \frac{[O]}{[R]}$$

$$E^{o'} = E^{\circ} + \frac{RT}{nF} \ln \prod \gamma^{\nu_i} = E^{o} + \frac{RT}{nF} \ln \frac{\gamma_O}{\gamma_R}$$

$$\mathbf{(R) = \gamma_R\ [R]}$$

$$\mathbf{(O) = \gamma_O\ [O]}$$

$$\mathbf{\frac{q}{E} = C}$$

$$\mathbf{i = nFAk^{o}\ [C_O(0,t)\ e^{-\alpha nf(E-E^{o'})} - C_R(0,t)\ e^{(1-\alpha)nf(E-E^{o'})}]}$$

i-E characteristics

$$i = FAk^{0}\left[C_0(0,t)e^{-\alpha f(E-E^{0'})} - C_R(0,t)e^{(1-\alpha)f(E-E^{0'})}\right]$$

Current-overpotential Eqn.

$$\frac{i}{i_0} = \frac{C_O(0,t)}{C_O^*}e^{-\alpha f\eta} - \frac{C_R(0,t)}{C_R^*}e^{(1-\alpha)f\eta}$$

Historically known as Butler-Volmer Eqn.

$$i = i_0\left[e^{-\alpha f\eta} - e^{(1-\alpha)f\eta}\right], \text{ when } C_O(0,t) = C_O^* \text{ and } C_R(0,t) = C_R^*$$

$$\mathbf{f = \frac{F}{RT}}$$

$$\mathbf{k_f = k^{o}\ e^{-\alpha nf(E-E^{\circ\prime})}}$$

$$\mathbf{k_r = k^{o}\ e^{(1-\alpha)\ nf(E-E^{\circ\prime})}}$$

$$i_0 = FAk^{0}C_0^{*}e^{-\alpha f(E_{eq}-E^{0'})}$$

$$i_0 = FAk^{0}C_O^{*(1-\alpha)}\ C_R^{*\alpha}$$

$$i_p = 0.4463\left(\frac{F^3}{RT}\right)^{1/2} n^{3/2}\, D_O^{\;1/2}\, C_O^*\, A\, v^{1/2}$$

$$i_p = 2.69 \times 10^5\; n^{3/2}\, D_O^{\;1/2}\, C_O^*\, A\, v^{1/2} \text{ at } 25\ ^\circ\text{C}$$

$$i_{ss} = 4nFD_O C_O^* r$$

$$i = \frac{nFAD_O^{1/2} C_O^*}{\pi^{1/2} t^{1/2}}$$

$$i = \frac{nFAD_O^{1/2} C_O^*}{\pi^{1/2} t^{1/2}} + \frac{nFAD_O C_O^*}{r}$$

$$i = \frac{n^2 F^2\, V\, v\, C_O^*}{RT} \frac{e^{-nf(E-E^{\circ\prime})}}{(1+e^{-nf(E-E^{\circ\prime})})^2}$$

$$i_p = \frac{n^2 F^2\, V\, v\, C_O^*}{4RT}$$

$$i_c = vC_{dl} = vC_{dl}^* A$$